Sequential Logic – Part 2
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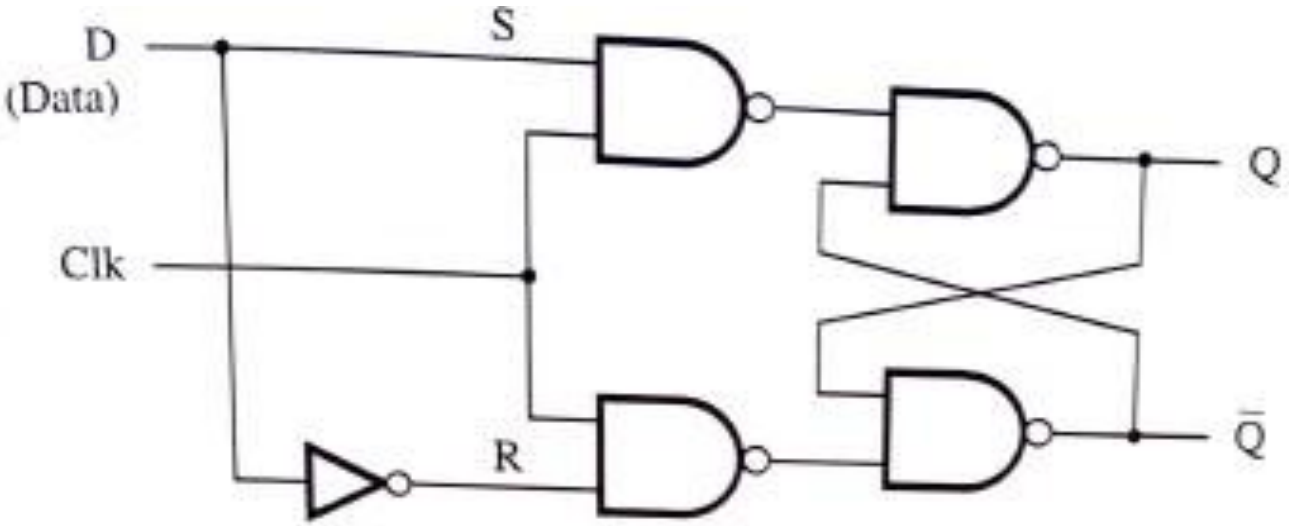
Review

- We have discussed basics of synchronous sequential logic circuits.
- We have then discussed about SR latch and how to control the output of the SR latch via a clock signal.
- We have also looked at handling an indeterminate situation in a gated SR latch by introducing a gated D latch.
- Then we have studied timing diagrams.

Review: What is past behaviour ?

- Recall we used the terms “past behaviour” when attempting to describe a sequential circuit (in the previous set of slides).
- We said outputs depend on past behaviour besides the present values of the inputs. We also indicated that this is a way to say how a sequential circuit is different from a combinational circuit.
- We will now see what past behaviour is with reference to the gated D latch introduced in part 1.

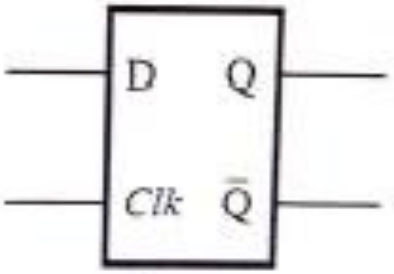
Review: Gated D latch



(a) Circuit

Clk	D	Q(t + 1)
0	x	Q(t)
1	0	0
1	1	1

(b) Truth table



(c) Graphical symbol

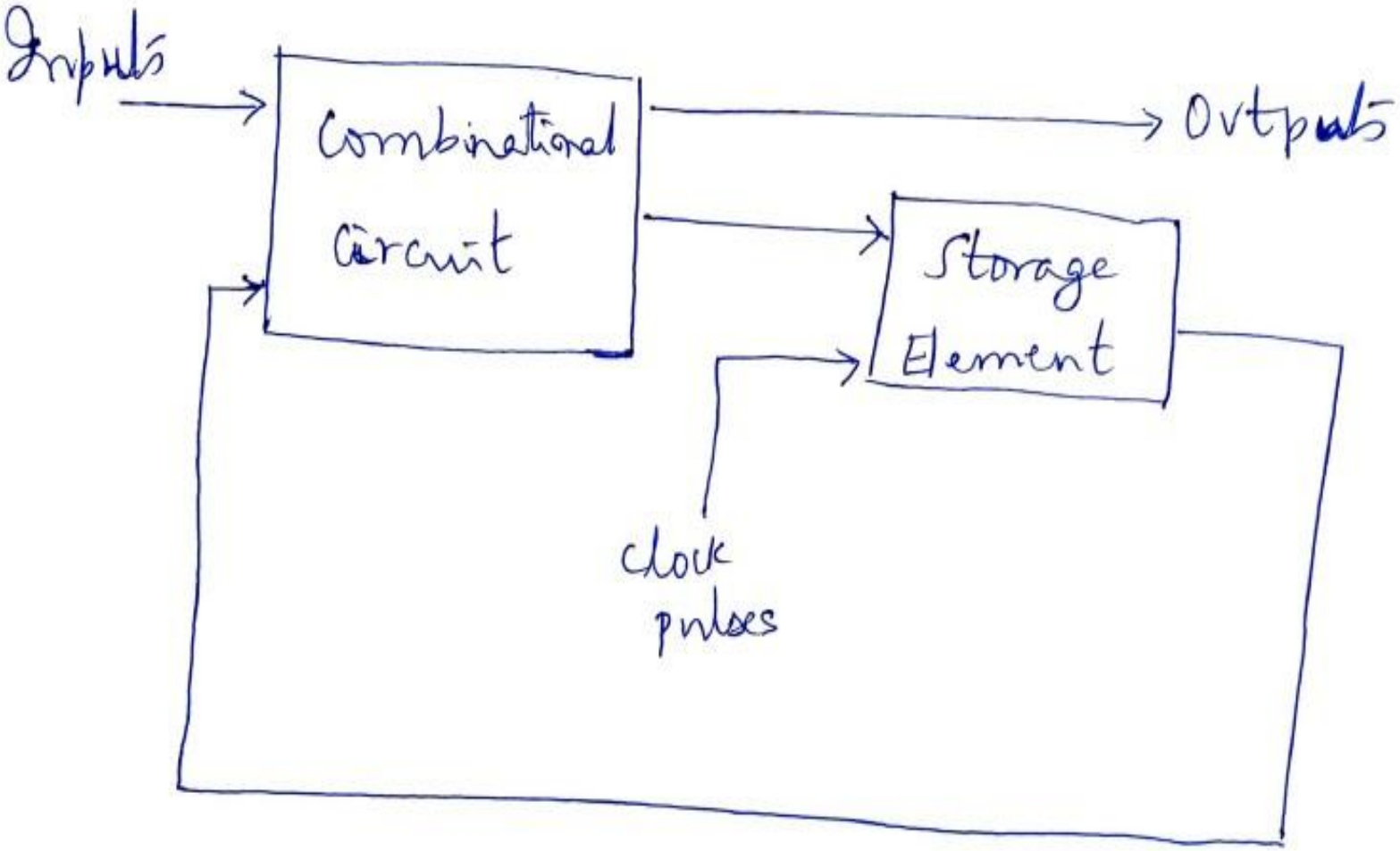
Consider the truth table ..

- The first line of the truth table for the gated D latch says when Clk= 0 and D = x (0 or 1), the output Q(t+ 1) is simply Q(t). If Q(t) had been 0, then Q(t+ 1) would also be 0. Similarly, if Q(t) had been 1, Q(t + 1) would also be 1.
- Q(t) could have been 1 because of D = 1 and Clk=1 applied in the past.
- We will now move on to other aspects of latches.

Are latches adequate for applications ?

- The SR latch and D latch have interesting properties. The gated versions of these are also promising.
- However, these latches are not generally used in isolation. They are used as part of a larger circuit that can include combinational logic elements besides sequential elements.
- A possible arrangement is shown next.

A clocked sequential circuit



Issues with latches

- Assume that the storage elements shown in the figure are latches.
- A latch output is, in principle, connected to another latch through a combinational circuit.
- State transitions (output changes) of a latch begin as soon as clock pulse changes to logic-1. The state of the latches can keep changing as long as the clock pulse remains at the active level so results may be unpredictable in general.

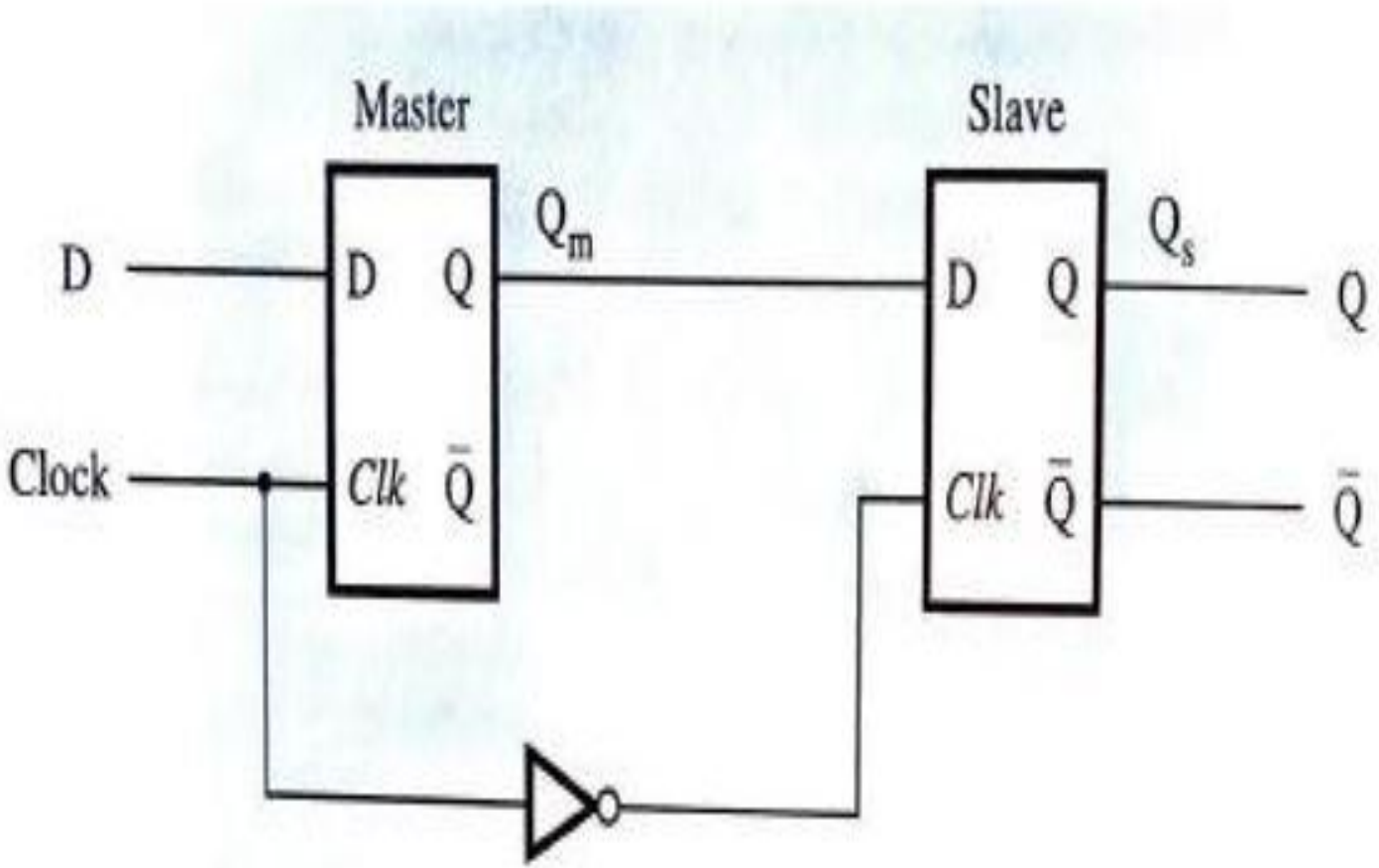
More on latches and alternatives

- We would often like to have storage elements that can change their states no more than once during a clock cycle.
- This calls for modification of the arrangement.
- Two approaches exist to modify a latch to form a flip-flop. The term flip-flop is used to denote a storage element that changes its output state at the edge of a controlling clock signal.

Modifications of a latch

- One approach is to use two latches in a certain configuration so that the (overall) output of the flip-flop is isolated from changes to the input to the flip-flop. This is called as a master-slave arrangement.
- Another is to design a flip-flop with fewer gates directly such that it triggers only during a signal transition (from 0 to 1) of the synchronizing clock signal.

Master-slave D flip-flop



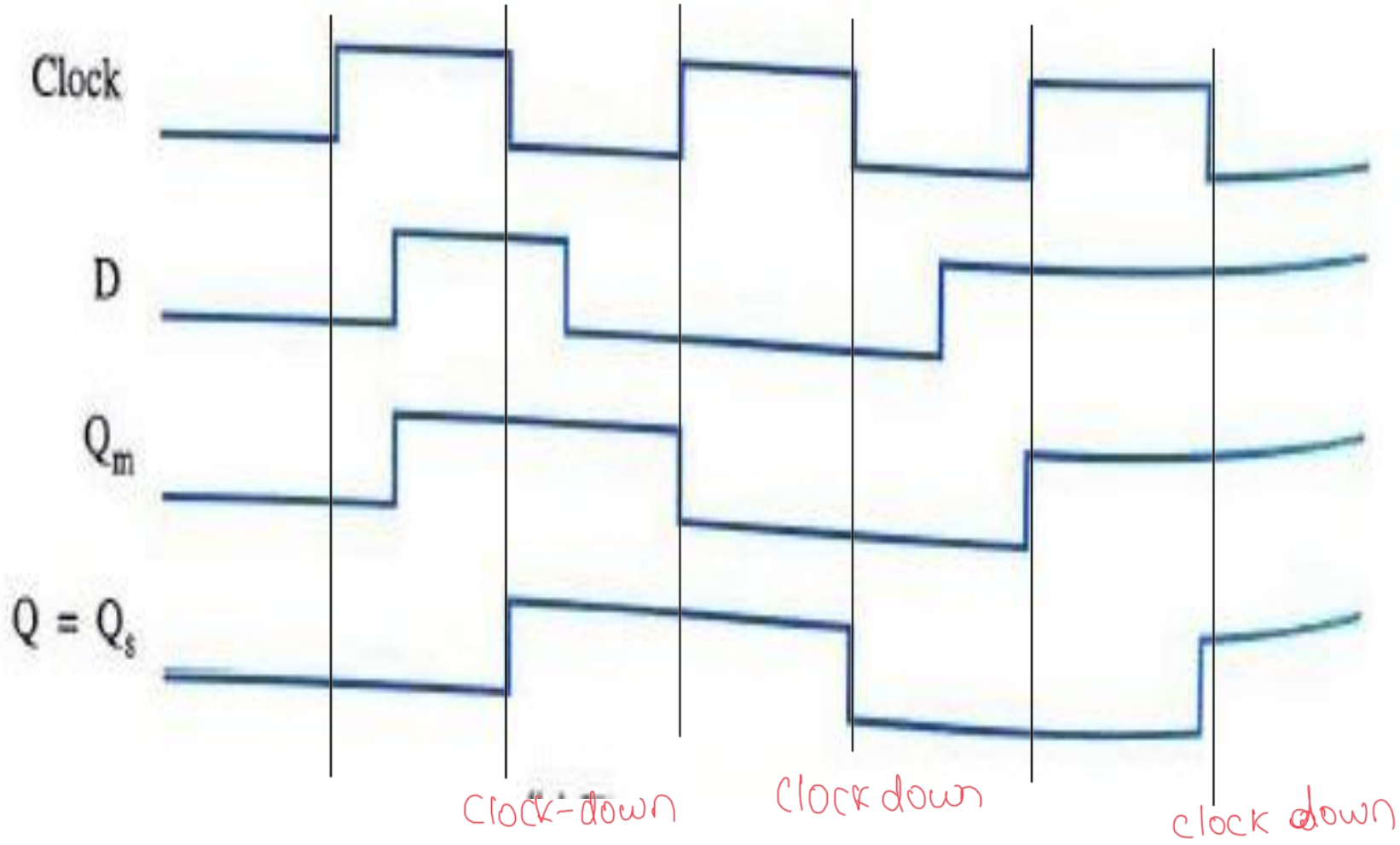
Explanation

- We have two gated D latches. The one on the left, called Master, changes its state when Clock = 1 while the second, called Slave, changes its state when Clock = 0 (notice an inverter in between).
- When Clock becomes 0, the master stage no longer responds to changes in D input. However, the slave stage responds to Q_m (output of master).
- Since Q_m does not change when Clock = 0, the slave stage can undergo at most one change of state during a cycle of Clock.

How does the circuit respond to the Clock edge ?

- From the observer's point of view, what is of interest is the "overall" output, namely Q_s : the master-slave circuit is seen to change state at the negative-going Clock edge (i.e., the edge where the Clock signal changes from 1 to 0).
- Irrespective of the number of changes in the D input to the master stage during one Clock cycle, the observer of Q_s will only see the change that corresponds to the D input at the negative Clock edge. The timing diagram helps understand better.

Timing Diagram



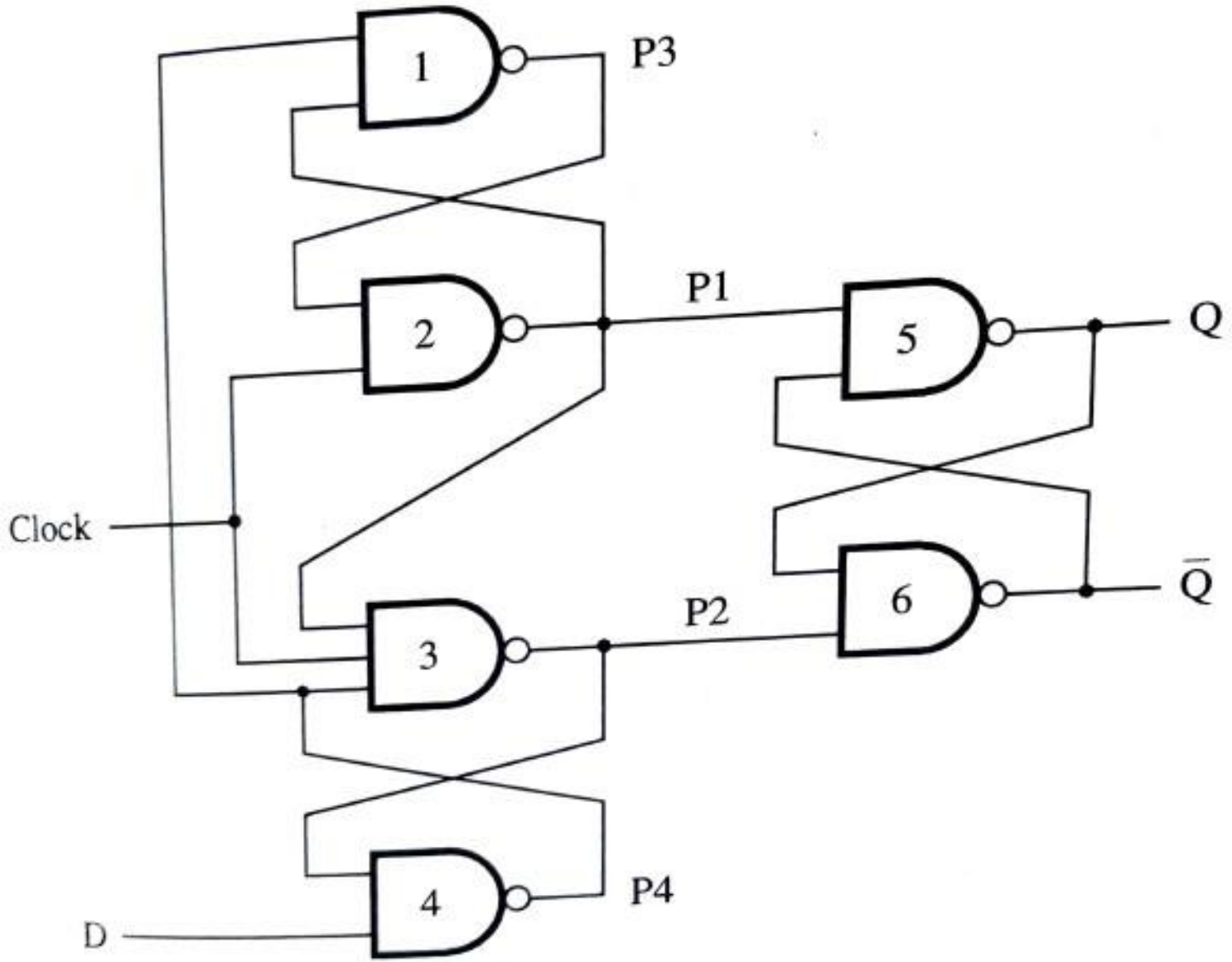
On the timing diagram

- The diagram shows that Q_m goes to 1 when D goes to 1 during the "level" phase of Clock.
- However, Q_s goes to 1 at the negative Clock edge. Of course, D applied to Master should be 1 for this to happen.
- Similarly, at a later point in the diagram, we can observe that Q_s goes to 0 at the negative Clock edge (and D is 0 at this instant).

Are there any issues with a master-slave D flip-flop ?

- While the master-slave arrangement works well, we need to examine the hardware required.
- As it stands, there are two latches and consequently, a fair number of gates. In particular, we have 4 NAND gates and 1 inverter per gated D latch so we require up to 8 NAND gates altogether (along with additional inverters).
- We can achieve the same functionality with fewer NAND gates as shown next.

Positive edge-triggered D flip-flop



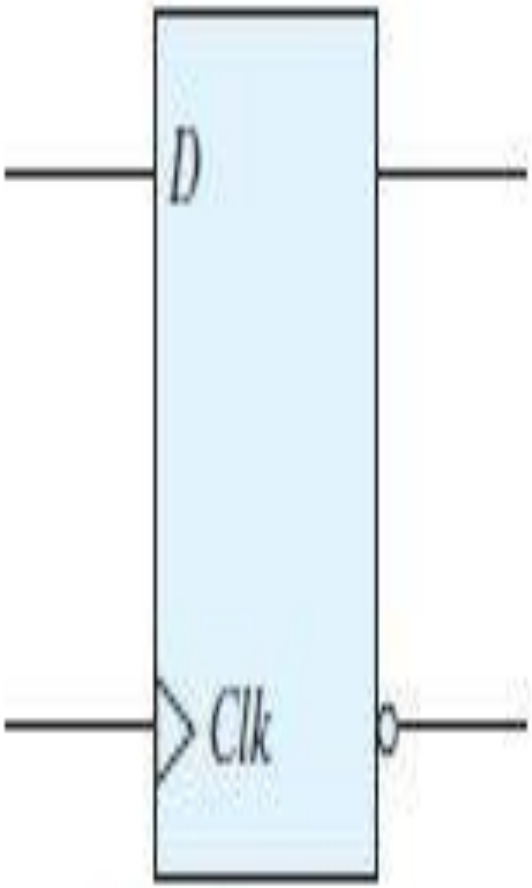
Explanation

- When Clock = 0, outputs of gates 2 and 3 are high. This means, P1 and P2 are 1.
- As a result, the state of the output latch comprising gates 5 and 6 is maintained.
- Now signal P3 is equal to D (verify this) and P4 is complement of D (i.e., Dbar).
- When Clock changes to 1: Values of P3 and P4 are transmitted through gates 2 and 3 causing P1 = Dbar and P2 = D. This leads to Q = D and Qbar = Dbar.

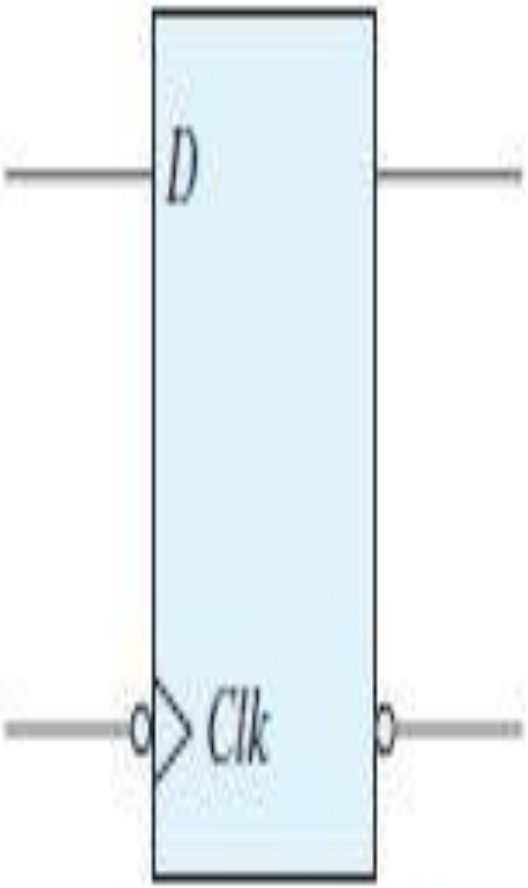
More on the positive edge triggered D flip-flop

- We want to ensure that after Clock changes to 1, further changes in D do not affect the output latch as long as Clock = 1. This involves considering both the following cases:
- First case: let D = 0 at the positive edge of the clock. Then P2 = 0 which will keep output of gate 4 equal to 1 (as long as Clock = 1) regardless of value of D
- Second case: let D = 1 at the positive edge of the clock. Then P1 = 0 forcing outputs of gates 1 and 3 to 1 regardless of input D.

Symbols for edge-triggered D flip-flops



(a) Positive-edge

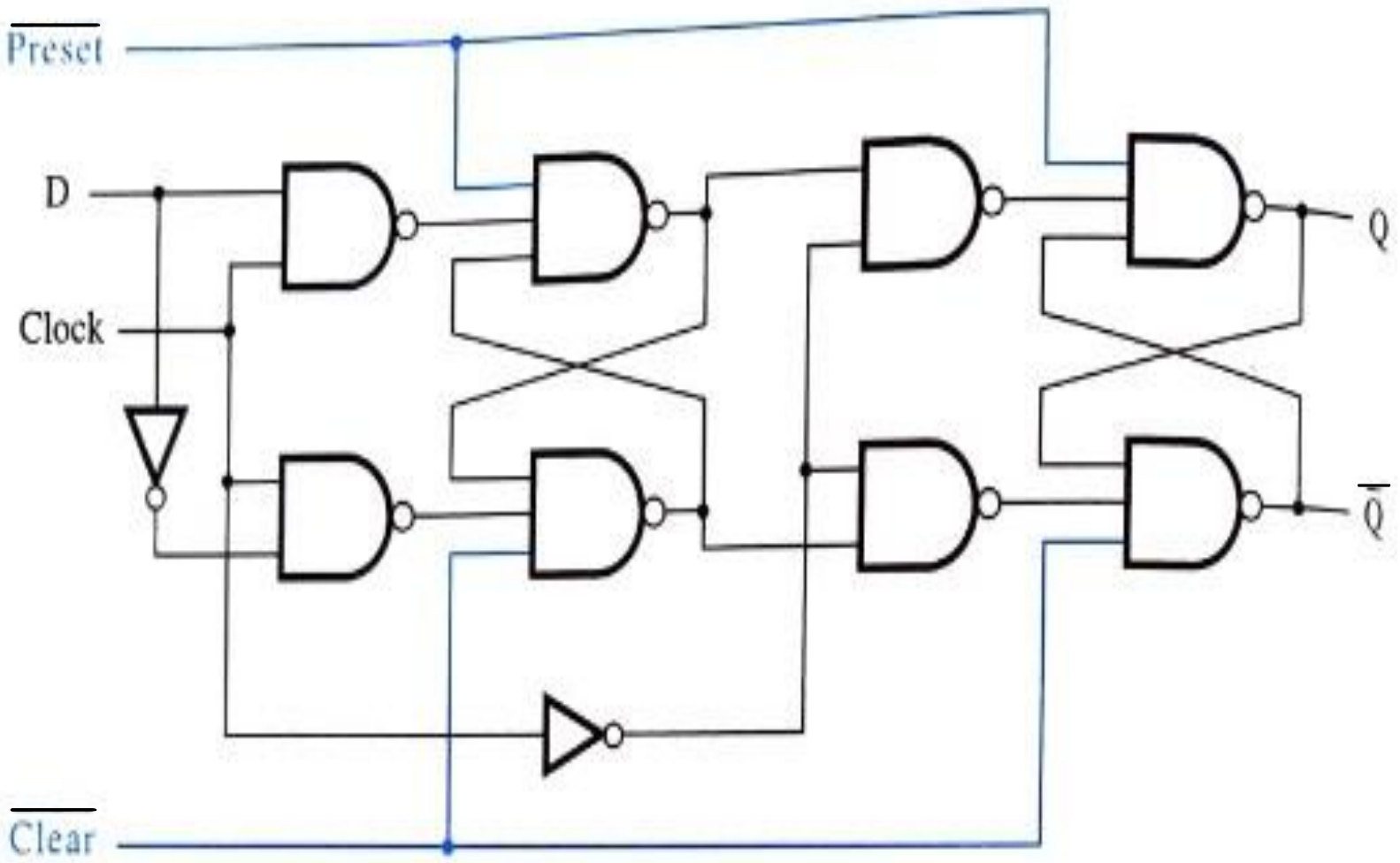


(a) Negative-edge

Adding a few extra inputs

- In some applications (such as counting), we will use multiple flip-flops and we need to have the capability to “preset” the count (to some value like 1) or “clear” the counter (i.e., we may have to force the counter to have an initial value of 0).
- For this purpose, we need to augment the flip-flop discussed earlier with a Preset input and a Clear input (besides the usual data and clock inputs).

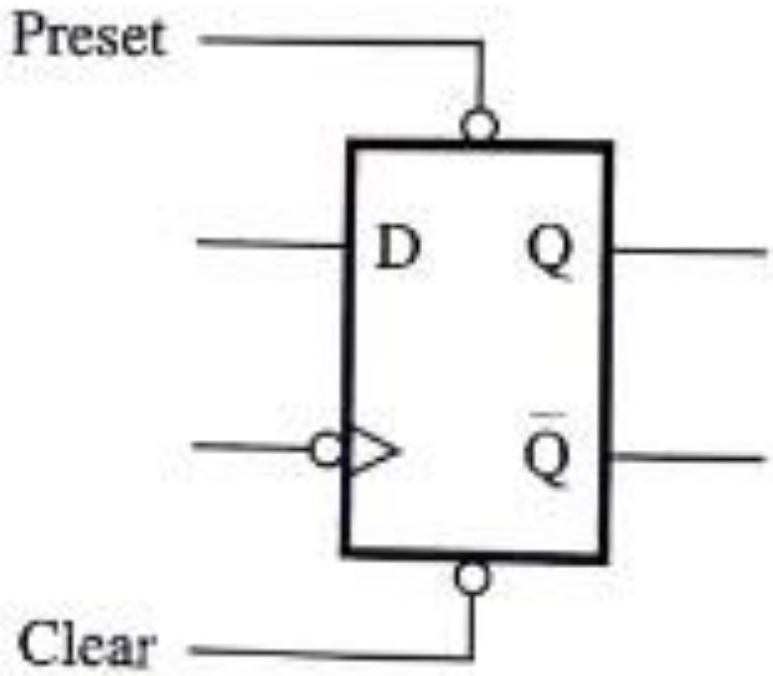
Flip-flop circuit with preset and clear



Explanation

- Placing a 0 on Clear input will force the flip-flop to have Q = 0 while Clear = 1 will have no effect on the NAND gates.
- Similarly, Preset = 0 forces the flip-flop into the state Q = 1 while Preset = 1 has no effect.
- To denote that the Clear and Preset inputs are active when their value is 0, we put an overbar line (on the names). Note that we should not try to force both Clear and Preset to 0 simultaneously.

Symbol with Preset and Clear



Textbook and Reference

- M. Mano and M. Ciletti, Digital Design, Pearson Education, 2018
- S. Brown and Z. Vranesic, Fundamentals of digital logic with VHDL design, McGraw Hill, 2017.